Federated LPV Identification for Heterogeneous Vehicles with Limited Local Speed Coverage

***Abstract*—Lateral vehicle control across varying speeds requires models that describe how steering responses change with longitudinal velocity. Yet each vehicle may supply identification data over only a narrow speed range. We develop a federated identification method that combines complementary observations from heterogeneous vehicles into group-specific linear-parameter-varying (LPV) models. Each vehicle estimates fixed-speed models from noisy state measurements and known applied steering using output-error identification. Clients then exchange additive statistics and model-assignment scores to learn group-specific LPV models without transmitting trajectories. Across ten independent simulated vehicle fleets, mean matrix error at locally unobserved speeds is 2.05%, compared with 3.94% for local identification with smoothness regularization and 4.48% for pooling all vehicles. Diagnostics show that accurate local estimates alone do not ensure informative groups. Closed-loop yaw-rate tracking uses linear-quadratic-integral (LQI) controllers designed from the identified models without further adaptation. Under noise-free state feedback, the proposed method satisfies the tested constraints and improves mean tracking over a fixed nominal controller. Additional gains over pooling all vehicles are small in absolute terms and depend on the recipient.**

***Index Terms*—federated identification, LPV systems, limited speed coverage, heterogeneous vehicles**

I. Introduction

Model-based vehicle control requires dynamics models that remain useful as operating conditions change. For lateral motion, longitudinal speed affects the relationship between steering, sideslip, and yaw rate. A model identified near one operating point may therefore be inadequate elsewhere. LPV models represent this dependence explicitly, making identification across the intended speed range a central requirement [1], [2].

Obtaining that coverage separately for every vehicle is a demanding calibration task. A vehicle may provide measurements over only a narrow part of its intended operating range. Those measurements constrain the observed response, but a flexible speed-dependent model can remain undetermined outside that region. Repeating measurements at the same speeds reduces noise without resolving missing scheduling directions. The resulting problem is therefore incomplete operating information as well as limited sample count.

Fleet collaboration offers a way to fill these gaps: one vehicle's unobserved speeds may be covered by others. Yet differences in mass, inertia, and tire properties make their measurements only approximately compatible. Pooling all vehicles can introduce bias, while separating them too aggressively can leave each group without sufficient speed coverage. What is needed is selective sharing that supplies missing operating information without overwhelming a recipient's dynamics with incompatible data. Crucially, the partners must be chosen from partial observations, before complete local models are available.

We address this need through federated LPV identification. Vehicles keep trajectories locally, exchange additive statistics, and evaluate candidate group models at their observed speeds. Alternating assignment and regression learns speed-dependent models over the target interval. The central question is whether groups inferred from partial observations can supply complementary speed information while describing their members' dynamics accurately. Figure 1 summarizes the procedure.

Existing work supplies important components of this approach. Local LPV identification constructs speed-dependent models from operating-point models [3]. Federated learning provides a framework for collaboration without centralizing records [4], and FedSysID characterizes the effect of dynamical heterogeneity on collaborative identification [5]. IFCA alternates cluster assignment and model optimization [6]. Clustered SysID applies this structure to LTI systems with identical dynamics within each true cluster [7]. IC-SYSID adds incremental clustering and stability promotion [8], while [9] uses uncertainty-aware grouping for vehicle controller design.

Our distinction is the joint requirement of similar dynamics and complementary speed observations. Each vehicle's observations leave its LPV regression underdetermined, and vehicles assigned to one group need not have identical dynamics. We adapt the established alternating clustering structure to this setting and make the following contributions:

1) We formulate LPV identification from limited local speed ranges in common physical-state coordinates, distinguishing missing speed information from uncertainty in the fixed-speed estimates.
2) We develop a federated procedure that combines local output-error estimates into group-specific LPV models. Additive statistics and local assignment scores allow vehicles to select sharing partners before each has identified its own model across the target speed range.
3) We evaluate identification at locally unobserved speeds against local regularization and global pooling on ten independent vehicle fleets. Closed-loop yaw-rate tracking with controllers designed from the learned models assesses whether improved identification also benefits control.

Sections II and III introduce the identification problem and federated method. Section IV describes the benchmark and evaluation protocol. Sections V and VI report identification

accuracy and closed-loop tracking. Section VII discusses the findings and limitations, and Section VIII concludes.

II. Identification Problem

Each vehicle needs a model across the target speed range but provides measurements at only a few speeds. First, we identify a fixed-speed model from each trajectory. Second, we combine these estimates into an LPV model, using other vehicles' observations to supply missing speed information.

A. Local identification from state measurements

Consider M vehicles indexed by $i \in \{1, \dots, M\}$ and the target speed interval $\mathcal{V} = [v_{\min}, v_{\max}]$, where $0 < v_{\min} < v_{\max}$. Vehicle i has state $\boldsymbol{x}_i = [\beta_i, r_i]^\top \in \mathbb{R}^2$, comprising sideslip angle β_i and yaw rate r_i, and scalar front-wheel steering input u_i. Both states are measured: $\boldsymbol{y}_{i,k} = \boldsymbol{x}_{i,k} + \boldsymbol{\nu}_{i,k}$, where k indexes samples and $\boldsymbol{\nu}_{i,k} \in \mathbb{R}^2$ is measurement noise. Applied steering and longitudinal speed are assumed known.

Vehicle i supplies trajectories at n_i observed speeds $\mathcal{V}_i = \{v_{i,1}, \dots, v_{i,n_i}\} \subset \mathcal{V}$. At each observed speed v, vehicle i records $N_i(v) + 1$ state measurements $\boldsymbol{y}_{i,0}, \dots, \boldsymbol{y}_{i,N_i(v)}$, sampled every T_s seconds, and the $N_i(v)$ applied steering inputs $u_{i,0}, \dots, u_{i,N_i(v)-1}$ between successive measurements. Section IV specifies how these data are collected.

Output-error (OE) identification estimates an affine state-space model by comparing its simulated response with the state measurements. Suppressing i and v for one vehicle–speed pair, write

$$\widehat{\boldsymbol{x}}_{k+1} = \boldsymbol{A}\widehat{\boldsymbol{x}}_k + \boldsymbol{B}u_k + \boldsymbol{c}, \qquad \widehat{\boldsymbol{x}}_0 = \boldsymbol{\zeta}, \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$, $\boldsymbol{B} \in \mathbb{R}^{2\times 1}$, $\boldsymbol{c} \in \mathbb{R}^2$, and $\boldsymbol{\zeta} \in \mathbb{R}^2$ are estimated jointly. For measurement-noise standard deviations $\sigma_\beta, \sigma_r > 0$, define $\boldsymbol{W} = \operatorname{diag}(\sigma_\beta^{-1}, \sigma_r^{-1})$. With $N = N_i(v)$, the OE objective is

$$\mathcal{L}(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{c}, \boldsymbol{\zeta}) = \frac{1}{2(N+1)} \sum_{k=0}^{N} \|\boldsymbol{W}(\widehat{\boldsymbol{x}}_k - \boldsymbol{y}_k)\|_2^2, \tag{2}$$

where $\|\cdot\|_2$ is the Euclidean norm and the simulated state follows (1) without resetting to measurements. The offset $\boldsymbol{c}$ avoids requiring a known equilibrium center, and $\boldsymbol{\zeta}$ describes the state at the start of the identification segment, after the vehicle has approached steady cornering.

The estimated matrices $\widehat{\boldsymbol{A}}_i(v)$ and $\widehat{\boldsymbol{B}}_i(v)$ form the row vector $\widetilde{\boldsymbol{m}}_i(v) \in \mathbb{R}^{1\times 6}$, ordered as $[\widehat{A}_{i,11}, \widehat{A}_{i,12}, \widehat{A}_{i,21}, \widehat{A}_{i,22}, \widehat{B}_{i,1}, \widehat{B}_{i,2}]$. They supply the scheduling dataset $\mathcal{D}_i = \{(v, \widetilde{\boldsymbol{m}}_i(v)) : v \in \mathcal{V}_i\}$. The estimated offsets and initial states remain local.

B. Missing speed information and group models

Let $\boldsymbol{\phi}(v) \in \mathbb{R}^L$ contain L linearly independent scheduling functions shared by all vehicles. Define $\boldsymbol{\Phi}_i \in \mathbb{R}^{n_i \times L}$ and $\boldsymbol{Y}_i \in \mathbb{R}^{n_i \times 6}$ with rows $\boldsymbol{\phi}(v_{i,\ell})^\top$ and $\widetilde{\boldsymbol{m}}_i(v_{i,\ell})$, respectively. The setting of interest has

$$\operatorname{rank} \boldsymbol{\Phi}_i < L. \tag{3}$$

Then some nonzero $\boldsymbol{U}_i \in \mathbb{R}^{L\times 6}$ satisfies $\boldsymbol{\Phi}_i \boldsymbol{U}_i = \boldsymbol{0}$. Adding $\boldsymbol{U}_i$ to a coefficient matrix leaves predictions at observed speeds unchanged but can change predictions elsewhere. Additional samples at the same speeds cannot remove this ambiguity. Exciting the dynamics at a fixed speed and observing their variation with speed are therefore separate requirements.

For K group models, let $g_i \in \{1, \dots, K\}$ assign vehicle i to a group with coefficients $\boldsymbol{\Theta}_{g_i} \in \mathbb{R}^{L\times 6}$. To describe approximation error, let $\boldsymbol{m}_i^\star(v)$ collect the vehicle's linearized dynamic matrices, defined for evaluation in Section IV. Write

$$\boldsymbol{m}_i^\star(v) = \boldsymbol{\phi}(v)^\top \boldsymbol{\Theta}_{g_i} + \boldsymbol{d}_i(v). \tag{4}$$

The row vector $\boldsymbol{d}_i(v) \in \mathbb{R}^{1\times 6}$ accounts for basis approximation and differences between a vehicle and its group model. Pooling can supply missing speed information, but this mismatch can reduce the benefit for individual vehicles. It is distinct from errors in the local OE estimates caused by measurement noise, finite-amplitude motion, and numerical optimization.

C. Federated identification task

Given the local datasets, we seek K, assignments g_i, and coefficients $\widehat{\boldsymbol{\Theta}}_g$ giving $\widehat{\boldsymbol{m}}_i(v) = \boldsymbol{\phi}(v)^\top \widehat{\boldsymbol{\Theta}}_{g_i}$ across $\mathcal{V}$. Clients exchange the statistics and scores defined below. True vehicle parameters, reference matrices, and physical family labels are excluded from fitting and grouping. Each vehicle contributes its observed speeds, and evaluation concerns predictions at its locally unobserved speeds. The target domain is the prescribed speed interval along a cornering condition, rather than the full nonlinear state–input domain.

III. Federated Learning of Group-Specific LPV Models

Using $\boldsymbol{Y}_i$ and $\boldsymbol{\Phi}_i$, we alternate group-model regression and vehicle assignment based on agreement at observed speeds. A complexity-penalized residual score selects the group count. We call this procedure *Grouped FL*. It selects sharing partners from partial observations; group coverage is assessed afterward rather than enforced.

A. Group regression from additive statistics

Given the current assignments, define the vehicles in group g as $\mathcal{I}_g = \{i : g_i = g\}$. Their coefficient matrix minimizes

$$\widehat{\boldsymbol{\Theta}}_g = \arg\min_{\boldsymbol{\Theta} \in \mathbb{R}^{L\times 6}} \left\{ \sum_{i \in \mathcal{I}_g} \|\boldsymbol{Y}_i - \boldsymbol{\Phi}_i \boldsymbol{\Theta}\|_F^2 + \rho \|\boldsymbol{\Theta}\|_F^2 \right\}, \tag{5}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\rho > 0$ is a common ridge parameter. Each speed–matrix target has equal weight. Client i computes $\boldsymbol{G}_i = \boldsymbol{\Phi}_i^\top \boldsymbol{\Phi}_i \in \mathbb{R}^{L\times L}$ and $\boldsymbol{H}_i = \boldsymbol{\Phi}_i^\top \boldsymbol{Y}_i \in \mathbb{R}^{L\times 6}$. The server sums these statistics and solves

$$\left(\sum_{i \in \mathcal{I}_g} \boldsymbol{G}_i + \rho \boldsymbol{I}_L \right) \widehat{\boldsymbol{\Theta}}_g = \sum_{i \in \mathcal{I}_g} \boldsymbol{H}_i, \tag{6}$$

where $\boldsymbol{I}_L$ is the identity matrix. For the same assignments, this is the centralized ridge-regression solution in exact arithmetic, without transmitting the local targets $\boldsymbol{Y}_i$.

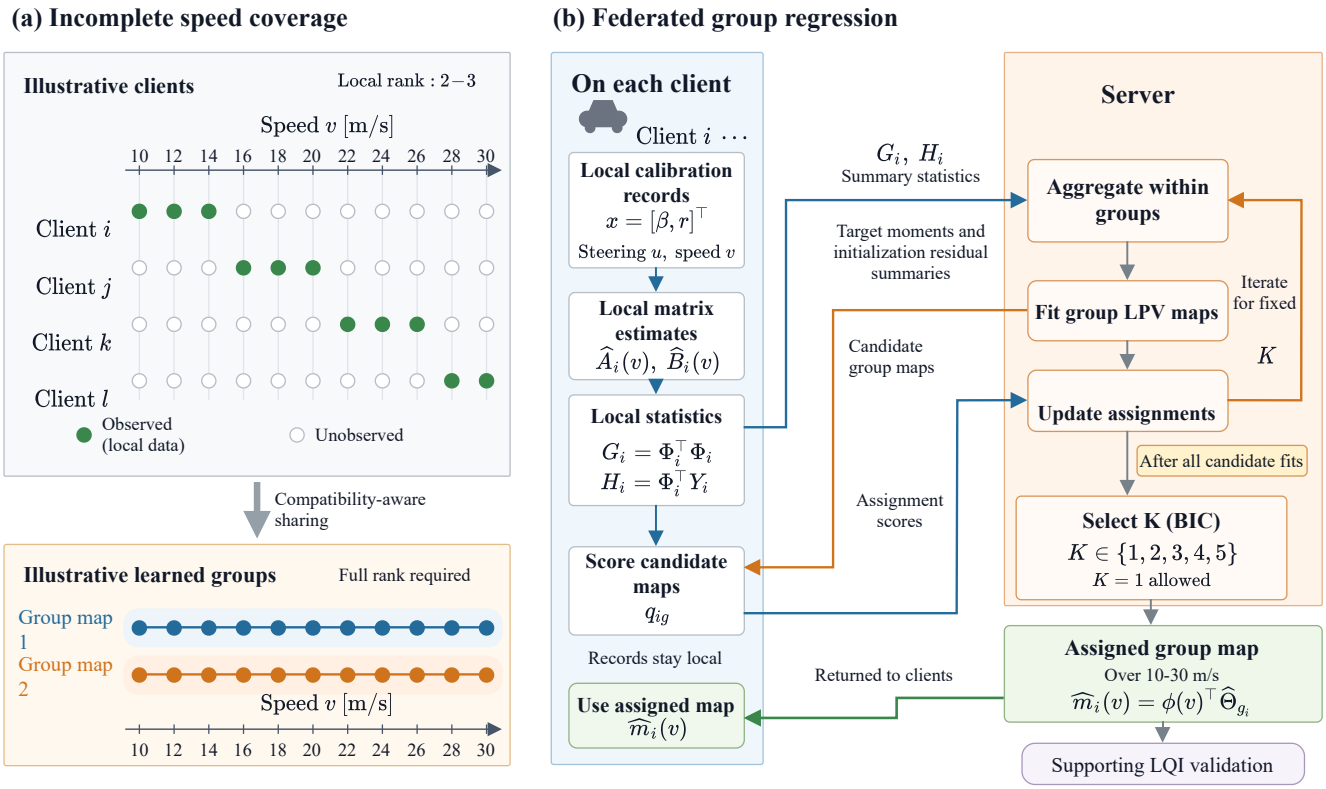


Fig. 1. Federated LPV identification from limited local speed ranges. (a) Illustrative vehicles and groups. Each local speed block supplies only two or three independent scheduling directions for a seven-function basis; combining observations can provide all seven. (b) Clients estimate fixed-speed models from state measurements and known applied steering. They exchange additive statistics and assignment scores, while trajectories remain local. Group regression and assignment alternate, and a BIC-based score selects the group count. The target speed range is 10–30 m/s along a prescribed cornering condition. Closed-loop tracking assesses the use of the learned models for controller design.

Let $\mathbf{\Phi}_g$ stack the designs of the group members. The unregularized fit is unique exactly when $\operatorname{rank} \mathbf{\Phi}_g = L$, since its Gram matrix is $\mathbf{\Phi}_g^\top \mathbf{\Phi}_g$. Thus complementary observations can resolve missing scheduling directions. Ridge regularization gives a unique solution even without full rank, but does not add information. Conditioning, local estimation error, and within-group mismatch also affect prediction accuracy.

B. *Vehicle assignment and group-count selection*

Assignment compares each candidate model with a vehicle's observed dynamics. A positive diagonal matrix $\boldsymbol{D} \in \mathbb{R}^{6\times 6}$ balances the six entries using their fleet-wide standard deviations, with a small common floor. It is computed from local target moments and fixed during the search. Unlike the OE weights $\boldsymbol{W}$, it describes variation across vehicles. After receiving the current group coefficients, client i returns

$$q_{ig} = \|(\boldsymbol{Y}_i - \mathbf{\Phi}_i \widehat{\mathbf{\Theta}}_g)\boldsymbol{D}^{-1}\|_F^2, \quad g = 1, \ldots, K. \tag{7}$$

The lowest score determines its assignment, with ties resolved by the lowest group index. Regression uses unscaled errors whereas assignment uses scaled errors. The alternating procedure is therefore heuristic, without a common-objective descent guarantee.

Initialization first fits one model to all vehicles, denoted *Global*, with coefficients $\widehat{\mathbf{\Theta}}_0$. Client i sends its mean residual

$$\boldsymbol{a}_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} \left[\widetilde{\boldsymbol{m}}_i(v_{i,\ell}) - \boldsymbol{\phi}(v_{i,\ell})^\top \widehat{\mathbf{\Theta}}_0 \right]. \tag{8}$$

The server standardizes residual entries across clients and applies K-means, using only observed-speed information. Constant entries use unit scale.

For residual sum $\mathrm{SSE} = \sum_i q_{i,g_i}$ and scalar target count $n = 6\sum_i n_i$, the Bayesian information criterion (BIC) based score is

$$\mathrm{BIC} = n \log\big(\max\{\mathrm{SSE}/n, 10^{-12}\}\big) + p_K \log n, \tag{9}$$

where $p_K = 6LK + K - 1$ counts model coefficients and a nominal mixture-weight term. This is a heuristic complexity penalty: mixture weights are not estimated, and OE target entries need not be independent. Algorithm 1 selects the lowest-scoring candidate, including *Global* at $K = 1$. A minimum group size limits small groups but does not guarantee sufficient speed coverage.

C. *Information exchanged*

Clients send $\boldsymbol{G}_i \in \mathbb{R}^{L\times L}$ and $\boldsymbol{H}_i \in \mathbb{R}^{L\times 6}$: 28 distinct symmetric entries and 42 entries, respectively, for $L = 7$.

Algorithm 1: Federated Learning of Group-Specific LPV Models

Input: M participating vehicles with local designs $\boldsymbol{\Phi}_i$ and OE targets $\boldsymbol{Y}_i$; common basis $\boldsymbol{\phi}(v) \in \mathbb{R}^L$; ridge ρ; maximum groups $K_{\max}$; restart budget N_{start}; iteration limit $T_{\max}$; minimum group size $M_{\min}$.

Output: Group count K, assignments g_i, and coefficients $\widehat{\boldsymbol{\Theta}}_g$, $g = 1, \dots, K$.

Initialize from locally observed dynamics

Clients: send $\boldsymbol{G}_i, \boldsymbol{H}_i$, counts, and target moments

Server: fit *Global* by (6); compute $\boldsymbol{D}$; send $\widehat{\boldsymbol{\Theta}}_0, \boldsymbol{D}$ to clients

Clients: return mean residuals $\boldsymbol{a}_i$ from (8)

Server: standardize residual entries across clients

Fit and compare candidate groupings

for $K = 1, \dots, K_{\max}$ **do**

  Use one start for $K = 1$ and N_{start} starts otherwise

  for *each start* **do**

    Set $g_i = 1$ for $K = 1$; otherwise initialize assignments by K-means

    for $t = 1, \dots, T_{\max}$ **do**

      Discard this start if any group has fewer than $M_{\min}$ vehicles

      Clients: send $\boldsymbol{G}_i, \boldsymbol{H}_i$ and group tags

      Server: fit groups by (6); send all coefficients to clients

      Clients: return scores q_{ig} from (7)

      Server: update assignments to the lowest-scoring group

      Break if assignments are unchanged

    Discard this start if any final group has fewer than $M_{\min}$ vehicles

    If assignments changed at the limit, refit final groups and recompute scores

    Save the candidate and its score (9)

Distribute selected models

Server: select the lowest-BIC valid candidate and send its assignments and group coefficients to the clients

Initialization adds target moments and residuals. Assignment exchanges K scores per client and K models of $6L$ coefficients. Message size is independent of trajectory length, while total communication depends on the search budget. Trajectories and targets remain local in the emulator, but shared statistics provide no formal privacy guarantee.

IV. Benchmark and Evaluation

We test whether selective sharing improves identification at locally unobserved speeds. Every comparison method receives the same local OE estimates, so differences reflect sharing and extrapolation rather than different local estimators. Settings were fixed after development on three pilot fleets and then applied to ten independently generated evaluation fleets.

A. Vehicle fleet and local speed coverage

The nonlinear single-track plant has state $\boldsymbol{x} = [\beta, r]^\top$, with sideslip measured at the center of mass, and steering input u. At fixed longitudinal speed v, its dynamics are

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{10}$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{11}$$

$$F_a = \mu F_{za} \tanh(C_a \alpha_a / (\mu F_{za})), \quad a \in \{f, r\}. \tag{12}$$

Here m is mass, I_z is yaw inertia, and l_f, l_r are the distances from the center of mass to the front/rear axles. For axle a, F_a, F_{za}, C_a, and α_a denote lateral force, normal load, cornering stiffness, and slip angle, with $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, and $F_{zr} = mgl_f/(l_f + l_r)$. Gravity is $g = 9.81$ m/s^2 and friction coefficient $\mu = 0.9$. Angles are in radians and yaw rate in rad/s.

TABLE I
Vehicle parameter centers used to generate the fleet. All clients have $l_f = 1.2$ m and $l_r = 1.6$ m.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

Each fleet contains 30 vehicles, ten from each parameter center in Table I. Mass and inertia have independent uniform scatter of $\pm 2.5\%$, axle stiffnesses $\pm 5\%$, and geometry is fixed. The target interval is $\mathcal{V} = [10, 30]$ m/s. Local speed blocks $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s are randomly assigned to eight, eight, seven, and seven vehicles, without using family labels. Thus 23 vehicles observe three speeds and seven observe two, giving $23 \cdot 3 + 7 \cdot 2 = 83$ vehicle–speed datasets per fleet.

B. Identification trajectories and local estimation

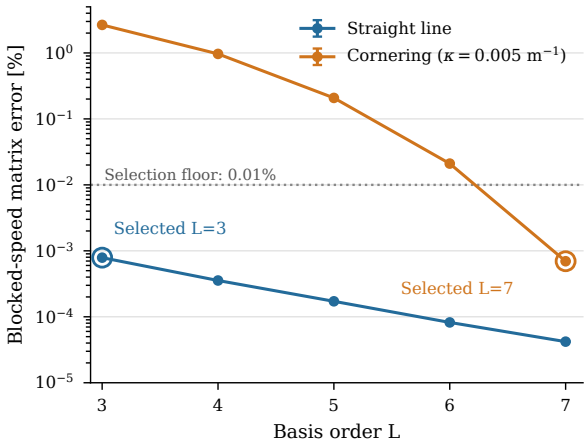
At each observed speed, a 44-s maneuver starts at $\beta = r = 0$. One nominal LTI model, linearized at straight driving and 20 m/s using the Nominal parameter center, supplies an LQI controller for all vehicles and speeds. The state–integral weights are $\boldsymbol{Q} = \mathrm{diag}(25, 50, 1000)$ and the steering weight is $R_u = 0.15$. The integral of measured yaw-rate error is bounded by ± 0.6 rad with conditional anti-windup. Steering is limited to $\pm 12^\circ$ and 30°/s.

During 0–4 s, the yaw-rate reference increases smoothly from zero to $v\kappa_0$, where $\kappa_0 = 0.005$ m^{-1} is the target curvature. The reference remains constant until 24 s, allowing the vehicle to approach steady cornering. Only the final 20 s are used for identification. In this segment, with local time $\tau = t - 24$ s, the reference is

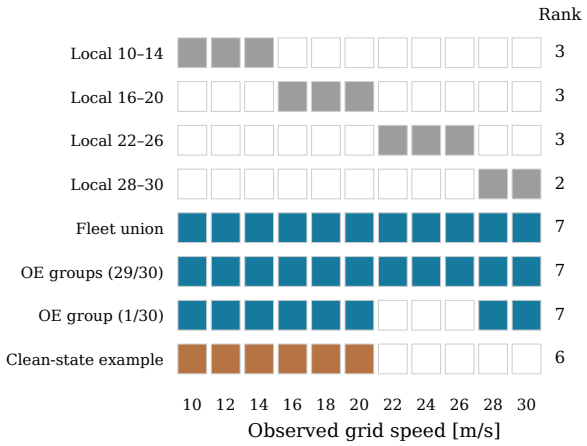
$$r^{\text{ref}}(\tau) = v\kappa_0 + \frac{a_{\text{exc}}}{v} h(\tau), \qquad a_{\text{exc}} = 0.3 \text{ m/s}^2. \tag{13}$$

The unit-peak waveform h sums sinusoids at 0.2, 0.5, 0.9, and 1.4 Hz with reproducibly randomized phases. Two-second sine-squared tapers switch the variation smoothly on and off. A separate maneuver with a 0.15–1.2 Hz chirp supports local-model diagnostics and trajectory checks without entering model estimation or selection. The LQI controller computes steering from the reference and measured states; sideslip follows the vehicle dynamics.

The 20-s identification segment provides 2001 state measurements and 2000 applied inputs at $T_s = 0.01$ s, giving $N_i(v) = 2000$. Measurement noise is independent zero-mean Gaussian across samples and components, with standard deviations 0.01° for sideslip and 0.03°/s for yaw rate. Both feedback and OE use these noisy state measurements. OE uses the known applied steering, without input measurement



(a) Development-only basis validation



(b) Scheduling information ($L = 7$)

Fig. 2. Scheduling representation and information coverage. (a) Earlier blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds. The 30 learned groups have two coverage patterns, both with rank seven. The brown row is a secondary clean-state group in evaluation fleet 5 with rank six. Full rank does not remove within-group mismatch or guarantee extrapolation accuracy.

noise. RK4 integrates the nonlinear dynamics with steering held within each step. OE minimizes (2) using four initializations and selects the converged solution with the lowest training loss.

Both maneuvers must satisfy steering bounds, $|\beta| \le 5°$, and $|a_y| \le 6$ m/s^2, where $a_y = (F_f + F_r)/m = v(\dot{\beta} + r)$ at fixed speed. Yaw-rate error and state drift must indicate steady cornering during the final 2 s before identification. Limiter activation is allowed because applied steering is recorded. Failed fits or checks invalidate the fleet condition without removing or replacing individual vehicles.

C. LPV representation and comparison methods

The common basis combines inverse-speed terms from the bicycle model with polynomial terms that allow additional variation around cornering:

$$\boldsymbol{b}(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \tag{14}$$

A fixed QR transformation on a dense speed grid defines the normalized basis $\boldsymbol{\phi}(v)$, improving conditioning without changing its span. Earlier blocked-speed validation on development reference matrices selected these seven functions for cornering and $[1, v^{-1}, v^{-2}]$ for straight driving (Fig. 2(a)).

The comparisons test three alternatives to selective sharing. *Restricted Local* asks whether a simpler speed dependence suffices: it fits the three straight-driving functions to each vehicle's own estimates with ridge 10^{-8}. *Global* tests whether sharing all observations is sufficient, fitting one seven-function model to the complete fleet. *Smooth Local* tests whether smoothness can replace collaboration, using the same seven-function basis but only the recipient's own observations:

$$\widehat{\boldsymbol{\Theta}}_i^{\mathrm{S}} = \arg\min_{\boldsymbol{\Theta}} \Big\{ \frac{\|\boldsymbol{Y}_i - \boldsymbol{\Phi}_i \boldsymbol{\Theta}\|_F^2}{n_i} + \alpha \|\boldsymbol{Q}_2 \boldsymbol{\Theta}\|_F^2 + \eta \|\boldsymbol{\Theta}\|_F^2 \Big\}. \tag{15}$$

Here $\boldsymbol{Q}_2$ implements the integrated squared second derivative of the matrix functions with respect to normalized speed z. It penalizes curvature while leaving affine functions unpenalized; $\eta = 10^{-16}$ is numerical ridge. Development selected $\alpha = 10^6$ as the largest penalty within one standard error of the best mean unobserved-speed error. All seven tested values $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, 10^6\}$ are also evaluated in a sensitivity analysis, without changing the primary setting.

Grouped FL follows Section III with $K_{\max} = 5$, $N_{\mathrm{start}} = 12$, $T_{\max} = 30$, $M_{\min} = 5$, and $\rho = 10^{-8} + 10^{-12}$. These fixed search budgets allow multiple initializations and early stopping at unchanged assignments.

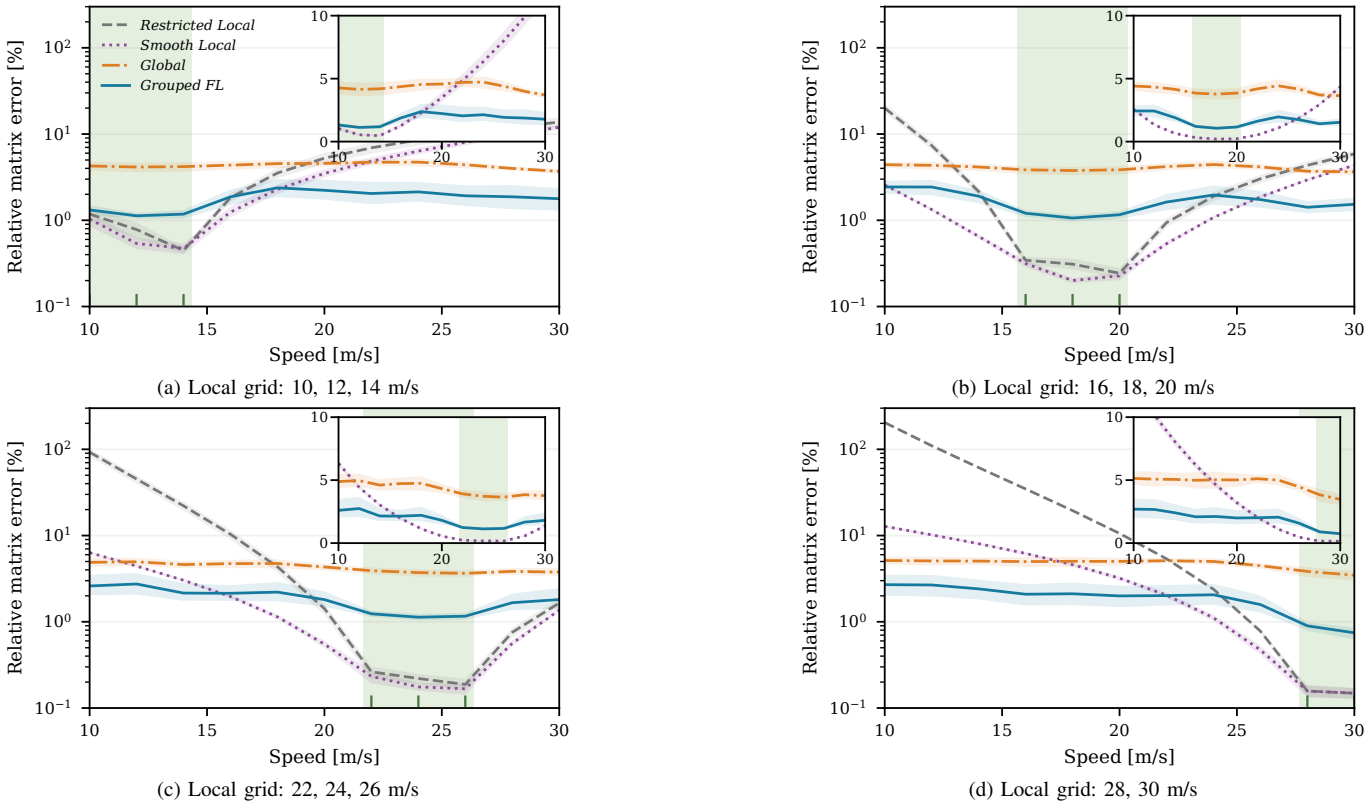
D. Evaluation metrics and statistical comparisons

Reference matrices are obtained from the known nonlinear simulation model. At each speed, we find steady cornering with yaw rate $r_i^\star = v\kappa_0$, linearize the dynamics with respect to state and steering, and discretize the resulting matrices at T_s using a zero-order hold. Their entries form $\boldsymbol{m}_i^\star(v)$ in the same order as the estimates. True recipient parameters are used for simulation and evaluation only, not for OE identification, grouping, or the nominal data-collection controller.

On $\mathcal{V}_{\mathrm{grid}} = \{10, 12, \ldots, 30\}$ m/s, define relative matrix error and mean unobserved-speed error for fleet s as

$$e_i(v) = \frac{\|\widehat{\boldsymbol{m}}_i(v) - \boldsymbol{m}_i^\star(v)\|_2}{\|\boldsymbol{m}_i^\star(v)\|_2}, \qquad E_s^{\mathrm{U}} = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{|\mathcal{U}_i|} \sum_{v \in \mathcal{U}_i} e_i(v), \tag{16}$$

where $\mathcal{U}_i = \mathcal{V}_{\mathrm{grid}} \setminus \mathcal{V}_i$ and $M = 30$. Speeds are averaged within vehicles, vehicles equally within fleets, and fleets equally across the experiment. Observed-speed and full-grid errors use $\mathcal{V}_i$ and $\mathcal{V}_{\mathrm{grid}}$ instead. Separate $\boldsymbol{A}$, $\boldsymbol{B}$, and entrywise errors test whether gains extend beyond the combined, state-unit-dependent metric.



(a) Local grid: 10, 12, 14 m/s

(b) Local grid: 16, 18, 20 m/s

(c) Local grid: 22, 24, 26 m/s

(d) Local grid: 28, 30 m/s

Fig. 3. Matrix error across the target speed range, grouped by locally observed speed block. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws, used descriptively. Insets show *Global*, *Grouped FL*, and *Smooth Local* on a linear 0–10% scale with identical averaging and bands. *Smooth Local* uses the curvature penalty selected before evaluation. These bands are distinct from the adjusted paired intervals for the primary endpoint.

For each primary baseline $b \in \{\textit{Smooth Local}, \textit{Global}\}$, use the paired fleet difference $\Delta_s^{(b)} = 100(E_s^{\mathrm{U},b} - E_s^{\mathrm{U,GF}})$ in percentage points. Here, GF denotes *Grouped FL*. Two Bonferroni-adjusted 97.5% two-sided paired-t intervals provide at least 95% simultaneous coverage under the paired-t assumptions. Fleets are the independent sampling units. Improvement requires both lower bounds above zero. Mean paired percentage gains are reported descriptively.

V. Identification Results

We first assess the local estimates and available speed information. We then compare prediction at unobserved speeds and examine the learned groups. Sensitivity checks test the local prior and error metric. Finally, coverage failures identify limits of sharing.

A. Local estimates and complementary speed observations

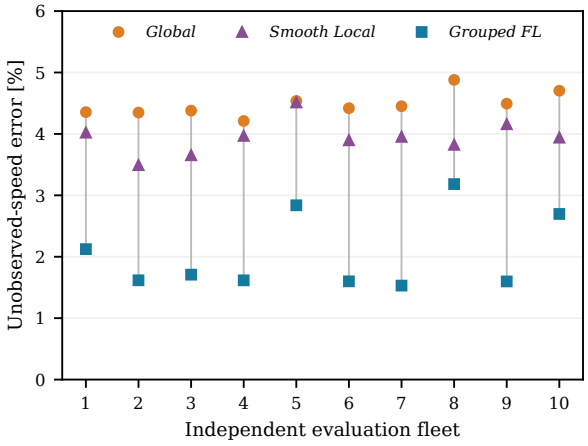
All $10 \cdot 83 = 830$ vehicle–speed OE fits converge, and all identification maneuvers satisfy the trajectory and settling checks. Local matrix error has median 0.247% and 95th percentile 1.206%. The steering limits introduced in Section IV-B are active during 6.37% of validation samples on average. The estimator uses the resulting applied input. A separate centralized solve with the same group assignments checks the federated regression against a tolerance of 10^{-8}. The two calculations agree within 5.12×10^{-14} in coefficients and 2.24×10^{-14} in predictions.

Accurate fixed-speed estimates do not resolve missing speed dependence. The four local blocks have scheduling ranks three, three, three, and two, whereas their union has rank seven. Figure 2(a) shows the development comparison used to select the basis. Figure 2(b) contrasts local speed observations with full-rank learned groups and a deficient group from a secondary diagnostic.

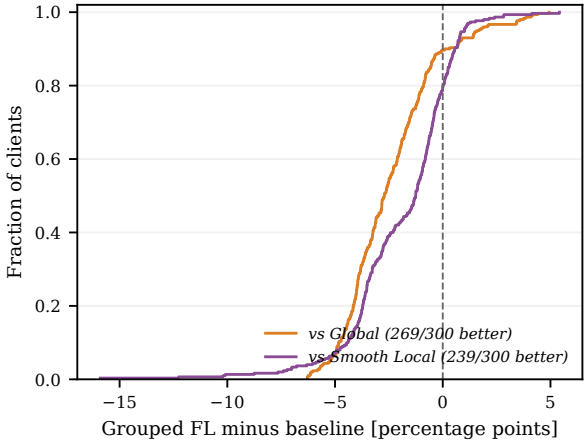
B. Identification accuracy at unobserved speeds

Table II summarizes the ten independent evaluation fleets. *Grouped FL* reduces mean unobserved-speed error to 2.05%, compared with 3.94% for *Smooth Local* and 4.48% for *Global*. The paired reductions are 1.89 percentage points against *Smooth Local*, with adjusted interval $[1.39, 2.39]$, and 2.43 points against *Global*, with interval $[2.02, 2.83]$. Both intervals exclude zero. Mean paired percentage gains are 48.0% and 54.5%, respectively.

Figure 3 groups vehicles by their locally observed speed block. Shading marks that block, and the curves show matrix error across the target speed grid. Local methods fit observed speeds more closely, whereas *Grouped FL* improves predictions outside those blocks. Figure 4(a) shows gains over both primary baselines in every fleet. Benefits vary among vehicles: 269/300 improve over *Global* and 239/300 over *Smooth Local*, while 31



(a) Paired primary results



(b) Client tradeoff (descriptive)

Fig. 4. Identification gains across evaluation fleets and individual vehicles. (a) Paired unobserved-speed errors; connectors join methods within each independent fleet. (b) Empirical cumulative distributions of client mean unobserved-speed error changes (*Grouped FL* minus baseline) in percentage points. Negative values indicate improvement. Client comparisons are descriptive; the 300 clients are nested within ten fleets.

TABLE II
MEAN RELATIVE MATRIX ERROR [%] ACROSS TEN INDEPENDENT EVALUATION FLEETS. ERROR AT LOCALLY UNOBSERVED SPEEDS IS THE PRIMARY ENDPOINT.

Method	Full grid	Observed	Unobserved
Restricted Local	16.02	0.38	20.44
Smooth Local	3.07	0.33	3.94
Global	4.34	3.87	4.48
Grouped FL	1.82	1.09	2.05

TABLE III
MATRIX-COMPONENT ERRORS AND FROZEN-SPEED STABILITY. $\boldsymbol{A}$ AND $\boldsymbol{B}$ COLUMNS ARE MEAN RELATIVE ERRORS [%] AT UNOBSERVED SPEEDS. THE LAST COLUMN COUNTS MATRICES WITH SPECTRAL RADIUS AT LEAST ONE OVER ALL 3300 CLIENT–GRID PAIRS.

Method	$\boldsymbol{A}$ error	$\boldsymbol{B}$ error	Unstable
Restricted Local	18.72	28.23	136
Smooth Local	2.73	10.98	272
Global	4.13	6.50	0
Grouped FL	1.75	3.68	0

and 61 worsen, respectively. Panel (b) shows these descriptive recipient-level differences.

C. What the learned groups capture

All ten evaluation fleets select $K = 3$. The resulting 30 group designs have rank seven and condition numbers 3.10–16.02 in the common normalized basis. Twenty-nine groups include observations at all eleven grid speeds. The remaining group lacks observations at 22–26 m/s but has data on both sides, with mean unobserved-speed error 1.32%. Thus full scheduling rank can emerge without observing every grid point; the grouping procedure did not enforce this coverage.

Physical family labels, inspected only after identification, give adjusted Rand indices from 0.518 to 1. The groups match the physical families in 4 of ten fleets. Other partitions mix families, so improved prediction does not require exact recovery of the generating classes.

D. Sensitivity to local regularization and error metric

We test whether the advantage depends on an unfavorable local smoothness penalty or on the combined matrix metric. *Grouped FL* improves every fleet–penalty comparison across the seven prespecified strengths (70/70). Selecting the best penalty retrospectively on the evaluation fleets gives $\alpha = 10^{-2}$ and reduces *Smooth Local* error from 3.94% to 3.88%. *Grouped FL* remains at 2.05%, with mean paired gain 47.3%. This sensitivity check does not replace the independently evaluated primary baseline setting.

Errors fall for both $\boldsymbol{A}$ and $\boldsymbol{B}$ (Table III) and, in aggregate absolute terms, for each of the six entries against both primary baselines. Stability was assessed after identification rather than imposed during estimation. All *Grouped FL* frozen-speed state matrices have spectral radius below one, with maximum 0.9765. Closed-loop tracking is evaluated separately in Section VI.

E. When group coverage limits prediction

Accurate local estimates need not produce informative groups. To test this, clean-state OLS uses exact simulator states and applied inputs from the same noisy-feedback trajectories, centered using the preceding quiet segment. Clean-state OLS reduces median local matrix error to 0.052%, yet in one fleet the selected partition contains a five-vehicle group observing only 10–20 m/s. Its design has rank six (Fig. 2(b)), and its unobserved-speed error is 38.76%, raising the fleet mean to 9.68%. Because both local targets and assignments change, this example identifies a coverage failure rather than isolating the effect of measurement noise.

Known-family pooling replaces learned assignments with physical labels to test whether family membership ensures useful sharing. It also fails when a physical family lacks observations at 10–14 m/s: two fleets have mean unobserved-

speed errors of 44.19% and 23.35%. Their group designs are full rank but have condition numbers above 3000 in the same basis. These cases explain why similarity alone is insufficient: sharing partners must also supply enough independent, well-conditioned speed information.

VI. Closed-Loop Yaw-Rate Tracking with Identified Models

We assess whether the identified models support useful controller design. The same 300 evaluation vehicles now perform tracking maneuvers without further model fitting or controller-weight tuning. *Nominal LTI* tests improvement over the data-collection controller, and *Global* tests the additional benefit of selective sharing.

A. Common controller design

The four identification methods supply scheduled LQI controllers. A privileged *Reference LPV* model fits the seven-function basis to each recipient's full reference grid. It provides a comparison, not a performance bound. *Nominal LTI* keeps the original 20-m/s gain and prefilter. Scheduled methods synthesize gains at 10, 12, …, 30 m/s.

At each node, the identified dynamic matrices define

$$\boldsymbol{A}_{\mathrm{a}} = \begin{bmatrix} \widehat{\boldsymbol{A}} & \boldsymbol{0} \\ T_s \boldsymbol{c}_r & 1 \end{bmatrix}, \quad \boldsymbol{B}_{\mathrm{a}} = \begin{bmatrix} \widehat{\boldsymbol{B}} \\ 0 \end{bmatrix}, \quad \boldsymbol{c}_r = [0, 1]. \tag{17}$$

Discrete Riccati synthesis uses the weights from Section IV-B and yields $\boldsymbol{K} = [\boldsymbol{K}_x, k_\xi] \in \mathbb{R}^{1\times 3}$ for state $[\beta, r, \xi]^\top$, where ξ integrates yaw-rate error. For zero-integral equilibrium of the fitted model at unit yaw reference, solve

$$(\boldsymbol{I}_2 - \widehat{\boldsymbol{A}})\boldsymbol{h} - \widehat{\boldsymbol{B}} w = \boldsymbol{0}, \quad \boldsymbol{c}_r \boldsymbol{h} = 1, \quad F = w + \boldsymbol{K}_x \boldsymbol{h}, \tag{18}$$

with $\boldsymbol{h} \in \mathbb{R}^2$ and scalar w. Then $\boldsymbol{x}_{\mathrm{eq}} = \boldsymbol{h}\bar{r}$ and $u_{\mathrm{eq}} = w\bar{r}$ satisfy the model and feedback law at constant reference $\bar{r}$ and $\xi = 0$. Gains $\boldsymbol{K}$ and scalar prefilters F are linearly interpolated between nodes. Synthesis uses the dynamic matrices, while OE offsets and initial states remain parameters of the local identification problems.

For yaw-rate reference r_k^{ref}, the command and candidate integral are

$$\begin{aligned} u_k^{\circ} &= -\boldsymbol{K}(v_k)[\beta_k, r_k, \xi_k]^\top + F(v_k) r_k^{\mathrm{ref}}, \\ \xi_{k+1}^{\mathrm{cand}} &= \mathrm{clip}_{[-0.6,0.6]}\{\xi_k + T_s(r_k - r_k^{\mathrm{ref}})\}. \end{aligned} \tag{19}$$

Steering is rate-limited relative to u_{k-1}, then amplitude-limited. Conditional anti-windup accepts the integral update only when the command is unsaturated or the increment counteracts limiting. Unlike identification, these control tests use noise-free state feedback to isolate the consequences of the learned models.

B. Tracking at *locally* unobserved speeds

Separate 12-s runs at $v = 13, 21, 29$ m/s test interpolation between synthesis nodes at speeds unobserved locally by every vehicle. Each starts at $\boldsymbol{x}_0 = \boldsymbol{0}$, $\xi_0 = 0$, $u_{-1} = 0$, with reference

$$r^{\mathrm{ref}}(t) = \frac{2.5}{v} \sin^3\!\left(\frac{\pi(t-2)}{4}\right), \qquad 2 \le t \le 10, \tag{20}$$

TABLE IV

Fixed-speed tracking: mean yaw-rate RMSE [deg/s] over ten reused fleets. Parentheses count constraint-violating runs out of 300 per speed. All finite runs contribute to the means.

Method	13 m/s	21 m/s	29 m/s
Nominal LTI	0.0460 (0)	0.0142 (0)	0.0123 (0)
Restricted Local	11.7827 (56)	0.1133 (1)	1.3302 (18)
Smooth Local	0.0343 (0)	0.5294 (6)	5.6664 (68)
Global	0.0216 (0)	0.0140 (0)	0.0087 (0)
Grouped FL	0.0207 (0)	0.0122 (0)	0.0068 (0)
Reference LPV	0.0204 (0)	0.0116 (0)	0.0064 (0)

and zero otherwise, where time is in seconds. Peak nominal lateral demand is 2.5 m/s^2. RK4 uses $T_s = 0.01$ s with steering held within each step. Common bounds are 12° steering, 120°/s steering rate, and 6° sideslip. The rate bound is higher than the 30°/s limit used for identification. Limiting and bound violations are recorded separately. Full-duration yaw RMSE includes finite responses that violate bounds, with equal vehicle and fleet weighting.

Nominal LTI, *Global*, *Grouped FL*, and *Reference LPV* satisfy the tested bounds without steering limiting. Their actual lateral acceleration and steering remain below 2.49 m/s^2 and 3.17°, respectively. *Grouped FL* approaches *Reference LPV* and improves mean tracking over *Nominal LTI* at every speed (Table IV). Absolute gains over *Global* are 0.00095–0.00194 deg/s, with improvements in 9/10 fleets at 13 m/s and all ten at 21 and 29 m/s.

Figure 5 shows two vehicles selected before this evaluation. Across the fleet, heavy-family vehicles benefit most consistently, while the handling-family mean slightly favors *Global* at 13 and 29 m/s. *Restricted Local* and *Smooth Local* violate bounds in 75/900 and 74/900 runs. These failed responses remain in the means and produce large RMSE values, so tracking error must be read together with the violation counts.

C. *Tracking under varying speed*

Two supporting 12-s stress tests prescribe $11 \to 29 \to 15$ and $29 \to 12 \to 28$ m/s, with peak nominal lateral demand up to 7.7 m/s^2. Both profiles are evaluated on all 300 vehicles, giving $10 \times 30 \times 2 = 600$ runs per method. *Grouped FL* satisfies the bounds in all 600 runs. Mean paired fleet-relative gains over *Nominal LTI* are 24.25% and 25.33%, versus 8.36% and 1.29% over *Global*. In these stress tests, all 100 heavy-family vehicles improve over *Global* and all 100 handling-family vehicles worsen. The control benefit therefore depends on the maneuver and recipient, even when fleet-average identification improves.

VII. Discussion

The results support sharing observations that are both dynamically similar and complementary in speed. Local accuracy alone does not resolve missing scheduling directions, and physical family membership does not ensure a well-conditioned group regression. The current assignment procedure does not enforce these information requirements. Recipient-level coverage and

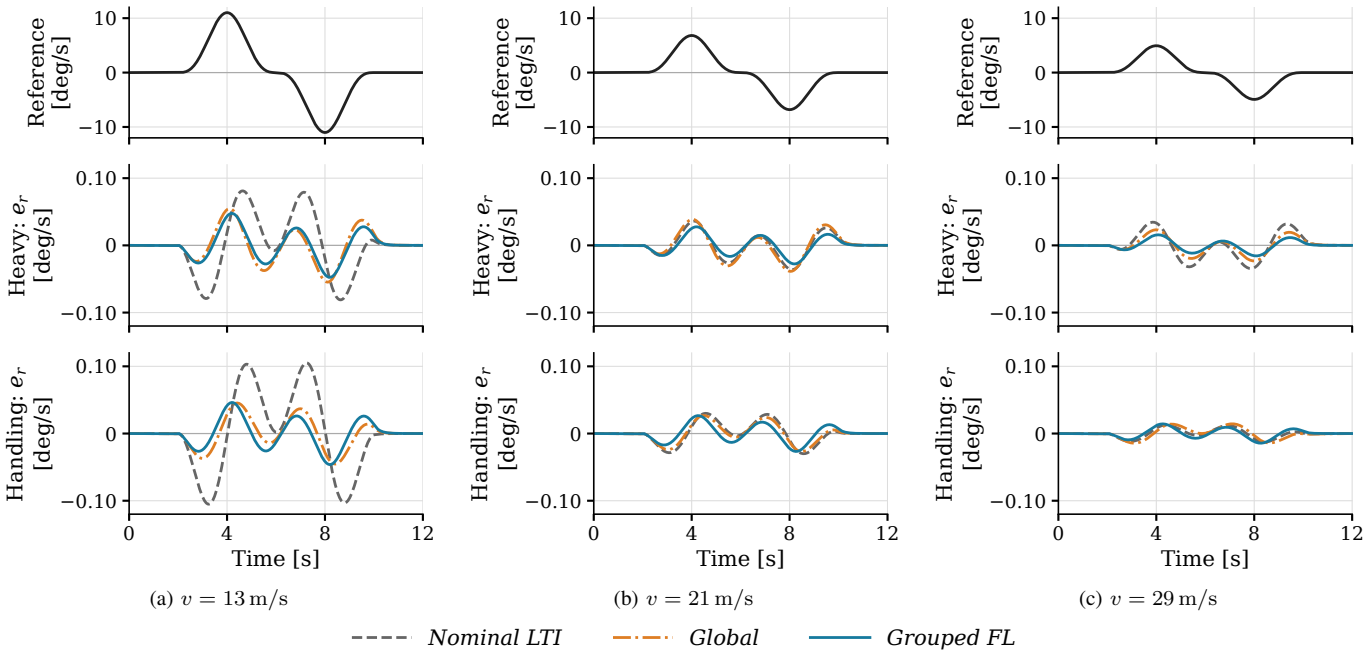


Fig. 5. Fixed-speed yaw-rate tracking at three speeds unobserved in local identification. Top: reference r_{ref}. Middle and bottom: errors $e_r = r - r_{\mathrm{ref}}$ for the same fixed heavy and handling recipients, respectively. All six error panels use identical limits. The reference has the same nominal lateral-acceleration demand at each speed.

conditioning checks are therefore important before using a shared model.

Closed-loop gains over *Global* are small and recipient-dependent. The tests use simulated vehicles, common controller weights, and noise-free state feedback. Prescribed-speed lateral dynamics omit longitudinal tire forces, braking load transfer, and combined slip, so the variable-speed tests are scheduling stress tests rather than validation of complete maneuver feasibility. Nonlinear and varying-speed stability, observer-based control, and full path tracking remain untested. All vehicles contribute local data, so transfer to new vehicles requires separate evaluation.

VIII. Conclusion

Limited local speed coverage leaves each vehicle's LPV model underdetermined. We address this by sharing complementary observations among vehicles with similar dynamics. Local OE identification supplies fixed-speed estimates, which are combined into group-specific LPV models without exchanging trajectories. Across ten independent simulated fleets, mean error at locally unobserved speeds is 2.05%, compared with 3.94% for *Smooth Local* and 4.48% for *Global*. Controllers designed from the learned models improve mean yaw-rate tracking over a fixed nominal controller and satisfy the tested constraints in all 1500 grouped-model runs. Additional tracking gains over global pooling are small and recipient-dependent. Future work should assess measured vehicle data and develop recipient-level checks for harmful transfer before controller deployment.

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